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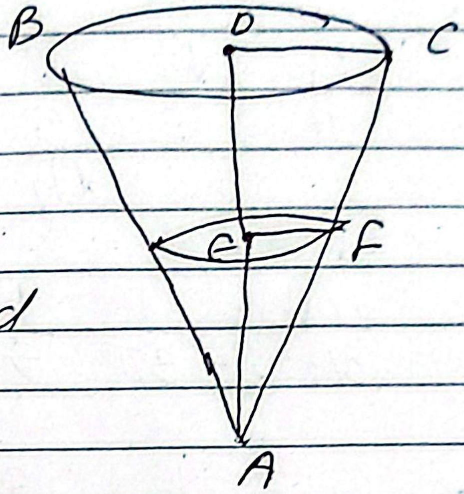
Set A

1) Soln:-

Let $\Delta ABC$ and $\Delta ADC$ be both triangle of the cone and water level. The height of the cone is 10 cm and radius is 3 cm. We know, $\Delta ABC$ and $\Delta ADC$ are ~~propa~~ similar triangle.




Now,

$$\frac{AE}{AD} = \frac{EF}{DC}$$

or, $$\frac{h}{10} = \frac{r}{3}$$

or, $3h = 10r$

or, $r = \frac{1}{3}h$

Now

We know,

Volume of cone $(V) = \frac{1}{3}\pi r^2 h$

$$V = \frac{1}{3}\pi \cdot \left(\frac{1}{3}h\right)^2 \cdot h$$

$$V = \frac{1}{3}\pi \cdot \frac{1}{9}h^2 \cdot h$$

$$V = \frac{1}{3}\pi \frac{1}{9}h^3 \quad \text{---- (i)}$$

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Differentiating eq$^n$(i) w.r.t, we get,

$$\frac{dV}{dt} = \frac{d\left(\frac{1}{3}\pi\frac{1}{9}h^3\right)}{dh} \cdot \frac{dh}{dt}$$

or $\frac{dV}{dt} = \frac{1}{3\pi} \cdot \frac{1}{9} \frac{dh^3}{dh} \cdot \frac{dh}{dt}$

or $2 = \frac{1}{27\pi} \cdot 3h^2 \frac{dh}{dt}$

or $\frac{2 \times 27\pi}{3h^2} = \frac{dh}{dt}$

or $\frac{54\pi}{3h^2} = \frac{dh}{dt}$

or ~~$18\pi = \frac{dh}{dt}$~~

or ~~$\frac{dh}{dt}$~~ $\frac{18\pi}{h^2} = \frac{dh}{dt}$

or $\frac{18\pi}{25} = \frac{dh}{dt}$

$\therefore \frac{dh}{dt} = 2.261946$

#

$$\frac{\frac{a}{b}}{\frac{c}{a}} = \frac{a}{b} \times \frac{a}{c}$$

2a) $\int_1^9 \frac{3\sqrt{x}-5}{\sqrt{x}} \cdot dx$

Soln:-

Let $I = \int_1^9 \frac{3\sqrt{x}-5}{\sqrt{x}} \cdot dx$

or $I = \int_1^9 \left(\frac{3\cancel{\sqrt{x}}}{\cancel{\sqrt{x}}} - \frac{5}{\sqrt{x}}\right) \cdot dx$

or $I = \int_1^9 3 \cdot dx - 5\int_1^9 \frac{1}{\sqrt{x}} \cdot dx$

or $I = 3\,[x]_1^9 - 5\int_1^9 x^{-1/2} \cdot dx$

or $I = 3[9-1] - 5\left[\frac{x^{-1/2+1}}{-1/2+1}\right]_1^9$

or $I = 3 \times 8 - 5\left[\frac{x^{1/2}}{\frac{1}{2}}\right]_1^9$

or $I = 24 - 5 \cdot \left[\frac{\sqrt{x}}{\frac{1}{2}}\right]_1^9$

or $I = 24 - 5 \cdot \left[\frac{\sqrt{x}}{1} \times \frac{2}{1}\right]_1^9$

or $I = 24 - 5 \times 2\left[\sqrt{9} - \sqrt{1}\right]$

$2x^2 - (12 - x^2)$
$2x^2 - 12 + x^2$
$3x^2 - 12$

or $I = 24 - 10[3-1]$
or $I = 24 - 10 \times 2$
or $I = 24 - 20$
$\therefore I = 4$ //

$-(3x^2-12)$
$-(12-3x$

b) Soln:-

$y = 2x^2$ and $y = 12x - x^2$

Find the intersection point,

$y = y$

or $2x^2 = 12x - x^2$
or $2x^2 + x^2 = 12x$
or $3x^2 - 12x = 0$
or $3x(x-4) = 0$
or $x = 0, x = 4$

To find ~~the~~ which equation is greater,
put $x = 0$

$y = 2 \times 0 = 0$
$y = 12 \times 0 - 0 = 0$

$\therefore$ we can take any function
Let $f(x) = 12x - x^2$ and $g(x) = 2x^2$

Now,

Area under curve (A) $= \left| \int_0^4 f(x) - g(x).\,dx \right|$

or $A = \left| \int_0^4 (12x - x^2 - 2x^2).\,dx \right|$

or $A = \left| \int_0^4 (12x - 3x^2).dx \right|$

or $A = \left| 12\int_0^4 x\,dx - 3\int_0^4 x^2\,dx \right|$

or $A = \left| 12\left[\frac{x^2}{2}\right]_0^4 - 3\left[\frac{x^3}{3}\right]_0^4 \right|$

or $A = \left| \frac{12}{2}\left[4^2 - 0^2\right] - \frac{\cancel{3}}{\cancel{3}}\left[4^3 - 0^3\right] \right|$

or $A = \left| 6 \times 16 - 64 \right|$

or $A = |96 - 64|$

or $A = 32$ #

3) Soln:-

Let f(x) be the probability function then, $\int_{-\infty}^{\infty} f(x).dx$ is always equal to 1.

$P(0 \le x \le 2) = \int_0^2 f(x).dx$

or $P(0 \le x \le 2) = \int_0^2 kx(2-x).dx$

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$$P(0 \le x \le 2) = \int_0^2 kx(2-x) \cdot dx$$

or $$1 = \int_0^2 kx(2-x) \cdot dx$$

or $$1 = \int_0^2 (2kx - kx^2) \cdot dx$$

or $$1 = 2k\int_0^2 x \cdot dx - k\int_0^2 x^2 \cdot dx$$

or $$1 = 2k\left[\frac{x^2}{2}\right]_0^2 - k\left[\frac{x^3}{3}\right]_0^2$$

or $$1 = \frac{\cancel{2}k}{\cancel{2}}\left[2^2 - 0^2\right] - \frac{k}{3}\left[2^3 - 0^3\right]$$

or $$1 = k\left[4-0\right] - \frac{k}{3}\left[8-0\right]$$

or $$1 = 4k - \frac{8k}{3}$$

or $$1 = \frac{4 \times 3k - 8k}{3}$$

or $$3 = 12k - 8k$$
or $$3 = 4k$$

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$\therefore K = \frac{3}{4}$

b) Soln:-

$P(3/2 \le x \le 3) = \int_{3/2}^{3} \frac{3}{4} x (2-x) \cdot dx$

or $P(3/2 \le x \le 3) = \int_{3/2}^{3} \frac{3x(2-x)}{4} \cdot dx$

or, $P(3/2 \le x \le 3) = \int_{3/2}^{3} \frac{6x - 3x^2}{4} \cdot dx$

or $P(3/2 \le x \le 3) = \frac{1}{4}\int_{3/2}^{3} (6x - 3x^2) \cdot dx$

or $P(3/2 \le x \le 3) = \frac{1}{4}\int_{3/2}^{3} 6x \cdot dx - 3\int_{3/2}^{3} x^2 \cdot dx$

or $P(3/2 \le x \le 3) = \frac{1}{4}\left[ 6 \cdot \left[\frac{x^2}{2}\right]_{3/2}^{3} - 3\left[\frac{x^3}{3}\right]_{3/2}^{3} \right]$

or $P(3/2 \le x \le 3) = \frac{1}{4}\left[ \frac{6}{2}\left[3^2 - (3/2)^2\right] - \frac{3}{3}\left[3^3 - (3/2)^3\right] \right]$

or $P(3/2 \le x \le 3) = \frac{1}{4}\left[ 3\left[9 - \frac{9}{4}\right] - \left[27 - \frac{27}{8}\right] \right]$

or $P(3/2 \le x \le 3) = \frac{1}{4}\left( 3\left(\frac{4 \times 9 - 9}{4}\right) - \left(\frac{27 \times 8 - 27}{8}\right) \right)$

or, $P(3/2 \le x \le 3) = \frac{1}{4}\left[3\times\frac{27}{4} - \frac{189}{8}\right]$

or, $P(3/2 \le x \le 3) = \frac{3\times 27}{4\times 4} - \frac{189}{8\times 4}$

or, $P(3/2 \le x \le 3) = \frac{81}{16} - \frac{189}{32}$

or, $P(3/2 \le x \le 3) = \frac{-27}{32}$

4) Sol$^{n}$:-

$T(x,y) = \frac{200}{x} + 2y + \frac{y^2}{50}$ --- (i)

Differentiating T w.r.t x we get,

$\frac{\partial T}{\partial x} = \frac{\partial(200x^{-1})}{\partial x} + \frac{\partial(2y)}{\partial x} + \frac{\partial\left(\frac{y^2}{50}\right)}{\partial x}$

$\frac{\partial T}{\partial x} = 200 \cdot \frac{\partial x^{-1}}{\partial x} + 0 + 0$ [we differentiate x and made y constant

$\frac{\partial T}{\partial x} = -1 \cdot x^{-1-1} \cdot 200$

$\frac{\partial T}{\partial x} = \frac{-200}{x^2}$

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b) Soln:-

$$\frac{\partial T}{\partial y} = \frac{\partial (200x^{-1})}{\partial y} + \frac{\partial (2y)}{\partial y} + \frac{\partial \left(\frac{y^2}{50}\right)}{\partial y}$$

$$\frac{\partial T}{\partial y} = 0 + 2 + \frac{1}{50}\frac{\partial y^2}{\partial y}$$ (we did x constant and differentiate w.r.t. to y)

$$\frac{\partial T}{\partial y} = 2 + \frac{1}{\cancel{50}_{25}} \cdot \cancel{2}y$$

$$\frac{\partial T}{\partial y} = 2 + \frac{y}{25}$$

c) Soln:-

To find the critical point, put $f_x = 0$

or, $\frac{-200}{x^2} = 0$

or, $\frac{0}{1} = \frac{-200}{x^2}$

or, $0 \cdot x^2 = -200$

or, $x^2 = \frac{-200}{0}$

No solution

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3) Soln:-

A = ∫∫

5) A square-based storage container without a lid is to have a volume of $8\ m^3$. Material for the base costs $20 per square meter, while for the sides costs $10 per square meter. Find the total cost for the least expensive container

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b) Soln:-

$P(3/2 \le x \le 3) = \int_{3/2}^{3} kx(2-x)$

$= \int_{3/2}^{3} \frac{3}{4} x(2-x) \cdot dx$

$= \int_{3/2}^{3} \frac{3x(2-x)}{4} \cdot dx$

$= \frac{1}{4} \int_{3/2}^{3} 6x - 3x^2 \cdot dx$

$= \frac{1}{4} \int_{3/2}^{3} 6x \cdot dx - \frac{1}{4} \int_{3/2}^{3} 3x^2 \cdot dx$

$= \frac{1}{\cancel{4}2} \times \cancel{6}3 \left[\frac{x^2}{2}\right]_{3/2}^{3} - \frac{1}{4} \times 3 \left[\frac{x^3}{3}\right]_{3/2}^{3}$

$= \frac{3}{4}\left[3^2 - \left(\frac{3}{2}\right)^2\right] - \frac{3}{4 \times 3}\left[3^3 - \left(\frac{3}{2}\right)^3\right]$

$= \frac{3}{4}\left[\frac{27}{4}\right] - \frac{3}{12}\left[27 - \frac{9}{4}\right]$

$= \frac{81}{16} - \frac{99}{61}$

$= \frac{4941 - 1584}{61 \times 16}$

⇒ 3.4395

$$\int_{3/2}^{3} \frac{3}{4}\, x(2-x)\,.dx$$

⇒ $$\int_{3/2}^{3} \frac{6x-3x^2}{4}\,.dx$$

⇒ $$\frac{1}{4}\left[\int_{3/2}^{3} 6x\,.dx - \int_{3/2}^{3} 3x^2\,.dx\right]$$

⇒ $$\frac{1}{4}\left[6\times\left[\frac{x^2}{2}\right]_{3/2}^{3} - 3\left[\frac{x^3}{3}\right]_{3/2}^{3}\right]$$

⇒ $$\frac{1}{4}\left\{\frac{\cancel{6}^{3}}{\cancel{2}_{1}}\left[3^2-\left(\frac{3}{2}\right)^2\right] - \frac{3}{3}\left[3^3-\left(\frac{3}{2}\right)^3\right]\right\}$$

⇒ $$\frac{1}{4}\left\{3\left(\frac{27}{4}\right)-\left(\frac{189}{8}\right)\right\}$$

⇒ $$\frac{1}{4}\left\{\frac{81}{4}-\frac{189}{8}\right\} = -0.84375$$

$$= \frac{81}{4\times4}-\frac{189}{4\times8}$$

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.b) $P(3/2 \le x \le 3) = \int_{3/2}^{3} \frac{3}{4} x(2-x)\,dx$

$= P(3/2 \le x \le 2) = \frac{3}{4}\int_{3/2}^{3} 2x - x^2\,dx$

$= \frac{3}{4}\left[ 2\left[\frac{x^2}{2}\right]_{3/2}^{3} - \left[\frac{x^3}{3}\right]_{3/2}^{3}\right]$

$= \frac{3}{4}\left\{ \frac{\cancel{2}}{\cancel{2}}\left[3^2 - \left(\frac{3}{2}\right)^2\right] - \frac{1}{3}\left[3^3 - \left(\frac{3}{2}\right)^3\right]\right\}$

$= \frac{3}{4}\left\{ \frac{27}{4} - \frac{1}{3}\left[\frac{189}{8}\right]\right\}$

$= \frac{3}{4}\left\{ \frac{27}{4} - \frac{63}{3\times 8}\right\}$

$= \frac{3}{4} \times \frac{33}{8}$

$= \frac{99}{32}$

$= \frac{32}{99}$

$= 0.32$